VIT BHOPAL www.vitbhopal.ac.in	**Forensic Chemistry and Applications**	**Course Type**	**LTP**
Course Code:	**CHY1007**	**Credits**	**4**
Prerequisite:	Chemistry of 12th standard or equivalent		

Course Objectives:

- To lay a suitable foundation of knowledge and experience to prepare the student to pursue an entry-level forensic scientist in the workforce.
- To get an insight of practical application of chemistry in forensic sciences.
- To show the importance of chemical principles, methods, and instrumentation for solving social issues through forensic investigations.

Course Outcomes (CO):

Students will be able to

CO1: To become better forensic investigators with precision.
CO2: To understand how analytical and spectroscopic methods are used for the analysis of forensic exhibits.
CO3: To know the chemical compositions of trace evidence at crime scene.
CO4: To understand the classification & analytical testing of illicit substances & their ill on human body.
CO5: To develop understanding about arson, explosives, classification of explosives and physical & chemical properties of explosives.

CO	Topics to be discussed	Hrs.
CO1	**Basics of forensic chemistry** Introduction to forensic chemistry - the flow of a forensic analysis, forensic chemistry today- Historical evidences – first forensic testimony. Metrology and measurements, Significant Figures, Rounding, and Uncertainty, Accuracy, Precision and Errors - types of analytical error, Sampling - sample preparation and screening. Expanding the scope of forensic chemistry. ***Tutorial**: Forensic analysis of popular case studies – Historical evidences.*	**8**
CO2	**Instrumentation – Forensic analysis** Chemical microscope, Spectroscopy – basics – Fourier Transform-Infrared (FT-IR), Ultraviolet-Visible (UV-Vis), Raman, mass spectrometry. Elemental analysis – X-ray fluorescence (XRF), powder X-ray techniques (PXRD). Gas Chromatography (GC), and High Performance Liquid Chromatography (HPLC) – SEM- EDAX. ***Tutorial**: Instrumentation analysis of forensic samples*	**9**
CO3	**Physical evidences: Inks, paints, paper and finger marks** Inks and paints: Composition, chemistry, forensic analysis - analytical methods – solubility test, TLC and chemical test (diphenylamine spot test for paints), aging and dating. Paper: Basics - forensic analysis - differential staining analysis - Herzberg's and Jenk's stain. Latent finger print analysis: Sources of latent fingermark residue - aqueous and lipid components, sources of compositional variation. Chemical processing of latent fingermarks – amino acid sensitive reagents – ninhydrin, 1,8-Diazafluoren-9-one, 1,2-indanedione and 1,2-indanedione and alternate reagents. Lipid sensitive reagents – Oil red O and Nile red. Other techniques - powder method. ***Tutorial**: Instrumental analysis of ink, paint, paper and finger print of forensic samples.*	**8**
CO4	**Drugs of abuse and alcohols** Drugs – overview and orientation, Classification and categories – by origin and function, general effect, use and by controlled substance Act and listed chemicals. History of drug regulations – Narcotics Drugs and Psychotropic Substance (NDPS) Act. Forensic drug	**9**

	analysis – Analytical approach to drug evidence, presumptive and confirmatory tests for acidic, basic, alkaloids and nonalkaloids drugs – color test, microcrystalline test and by TLC- Drugs effect in the human body. ***Tutorial**: Instrumental drug analysis and case study.*	
CO5	**Chemistry of Combustion and Arson and Explosives** Combustion and Arson – Combustion continuum, Fire and flames, arson and fire investigation, Accelerants and Incendiary Devices, Forensic Analysis of Fire Debris. Explosives – Explosions, detonations and explosives, Explosive power. Classification of explosives – low and high explosives, power index, physical and chemical properties of important explosives: black powder, perchlorate explosives, TNT, RDX, PETN, TATP and urea nitrate. Battery (secondary) explosions. Forensic examination of explosives – chemical analysis - Ignition susceptibility test, color/spot tests; separation methods – TLC ***Tutorial**: Instrumental analysis for bomb blast and battery explosion.*	**9**
CO5	**Contemporary topics:** Challenges in forensic investigation – Lecture from forensic experts.	**2**
	Total Lecture hours:	**45**

List of Experiments

1. To determine the accuracy & precision of the given data in hand.
2. To determine the presence of functional groups by the given UV-Visible spectra of a unknown food dye.
3. To elucidate the structure & determine the presence of different functional groups present in the given IR spectra of unknown compound.
4. To separate the coloured components present in the mixture of inks by ascending paper chromatography and determine their Rf values.
5. To prepare, study and graph fingerprint patterns.
6. To identify and analyze the fingermarks from various objects using Powdered Technique.
7. Estimation of Fe(II) in the debris sample by Redox Potentiometery.
8. Calorimetric estimation of poison in the given unknown sports drink sample.
9. Determination of pH of soil from the crime scene.
10. To estimate the amount of poison in the given unknown sample by Iodometric method
11. Estimation of Fe(II) in the debris sample by Cerimetry.

Text Books:

1. Suzanne Bell, "Forensic Chemistry" 1st ed., 2006 and 2nd ed., 2014 Pearson publisher.
2. Jay A. Siegel, "Forensic Chemistry: Fundamental and Applications"1st ed., 2016, John Wiely& Sons.
3. Suzanne Bell, "Forensic Chemistry" Annual Review of Analytical Chemistry, 2009, 2:297-319. DOI: 10.1146/annurev-anchem-060908-155251.

Reference books:

1. James E. Girard, "Criminalistics: Forensic Science and Crime" 2nd ed., 2011, Jones and Barlett publishers.
2. Giriraj Shah, "Encyclopaedia of Narcotic Drugs And Psychotropic Substances (3 Vols.)" 1st ed., 2004, Gyan Publishing House.
3. Practical aspects of forensic chemistry – A handbook of practical forensic chemistry, toxicology, narcotics and explosives
4. Anil Kr. Teotia, Rishi Pal, "Practical aspects of forensic chemistry – A handbook of practical forensic chemistry, toxicology, narcotics and explosives" 1st ed., 2013, Selective and Scientific Books.
5. Thomas Reddy, "Linden's Handbook of Batteries" 4th Ed., 2010, McGraw-Hill Professional

Recommendation by the Board of Studies on	
Approval by Academic council on:	
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